

Chemistry

Concise Lecture Notes

Some Basic Concepts of Chemistry

Key topics covered in the Class:

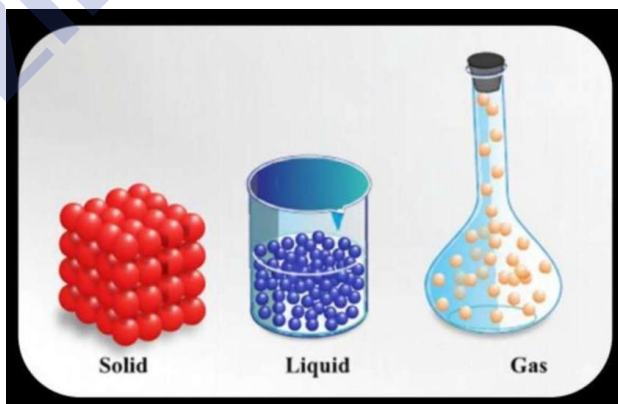
- Matter Classification (Physical & Chemical)
- Elements, Compounds, and Mixtures
- Laws of Chemical Combination
- Atoms, Molecules, Ions, Isotopes, Isobars, Isotones, Isodiaphers
- Mole Concept, Atomic/Molecular Mass, and Avogadro's Hypothesis
- Percentage Composition, Empirical/Molecular Formula, and Stoichiometry

Definition of Matter

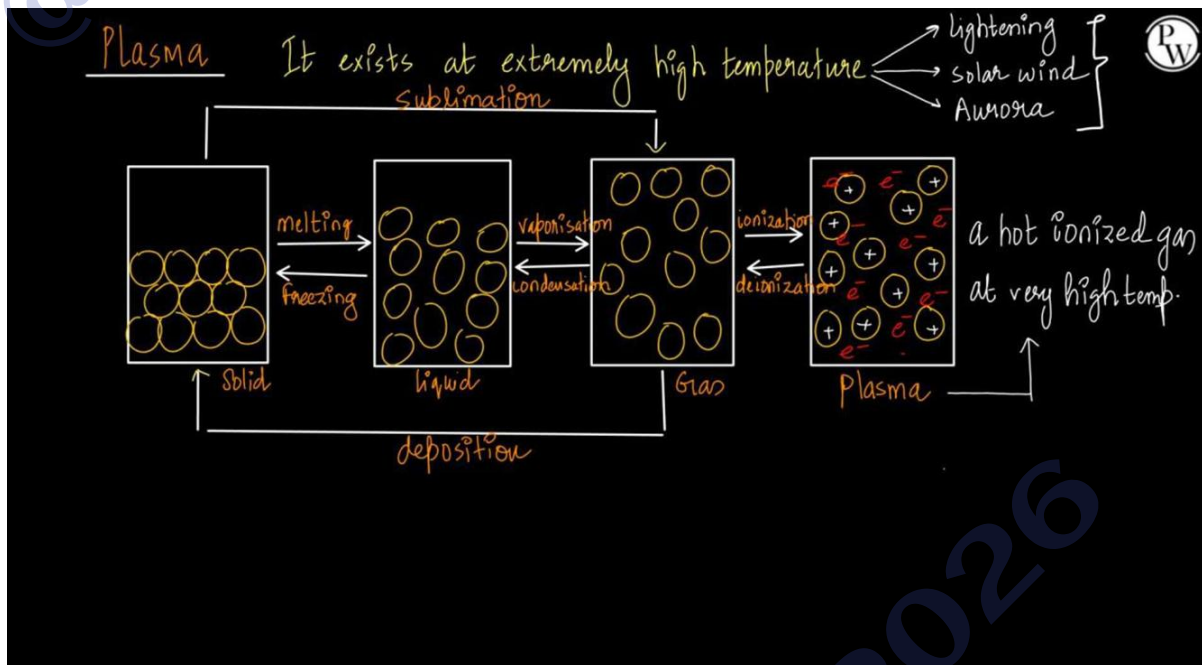
- Matter is anything which has mass and occupies space.
- **Non-Matter** refers to the things that do not occupy space, do not have mass, or consist of particles (e.g., emotions, time, light).

Physical Classification of Matter

- Matter exists mainly in three states: Solid, Liquid, and Gas.
- **Solid**: Has fixed shape and fixed volume; intermolecular forces are maximum; non-compressible.
- **Liquid**: Does not have a fixed shape but has a fixed volume; intermolecular forces are moderate; non-compressible.
- **Gas**: Neither has a fixed shape nor a fixed volume; intermolecular forces are minimum; compressible.

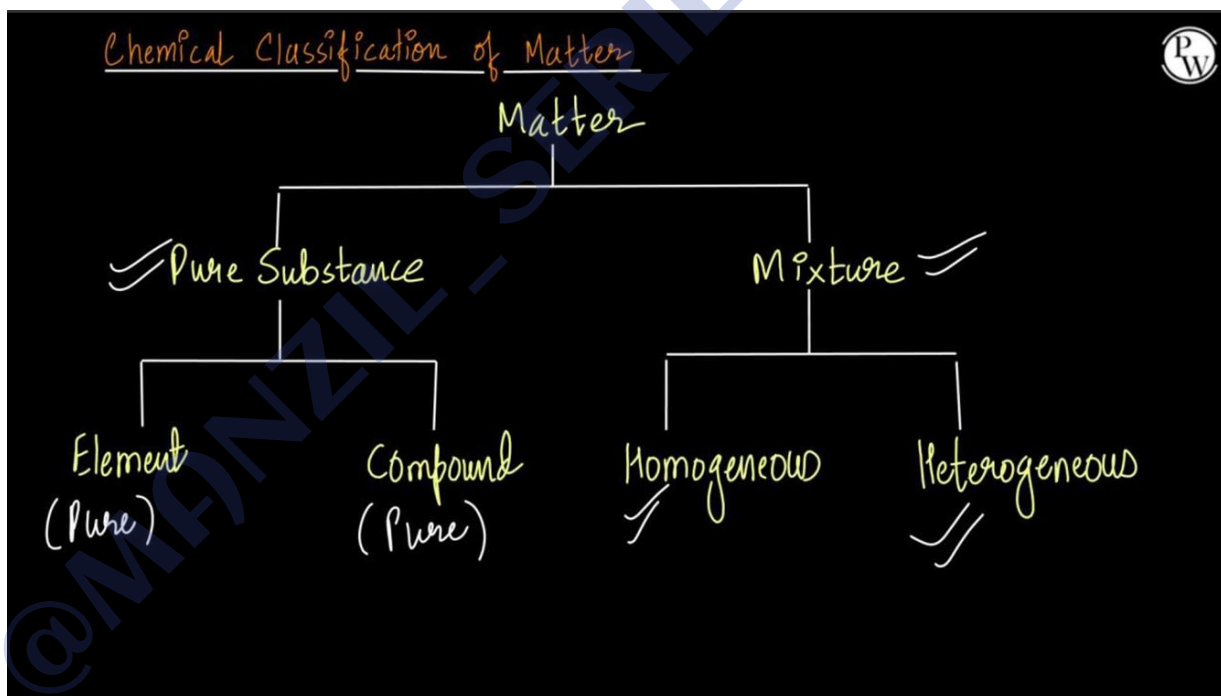


- Two additional states exist: **Plasma** (a hot ionized gas at very high temperature) and **Bose-Einstein Condensate (BEC)** (exists at very low temperature, almost 0 K or -273°C).



Chemical Classification of Matter

- Matter is classified into **Pure Substances** (Elements and Compounds) and **Mixtures** (Homogeneous and Heterogeneous).



Pure Substance

- A material containing only one type of substance.
- It cannot be separated into simpler substances by physical methods.

Element

- A pure substance containing one type of atom (e.g., Fe, O₂, P₄).

Compound

- A pure substance containing more than one kind of element which combine together in a fixed mass ratio (e.g., H₂O, HCl, NH₃).
- The property of the compound is entirely different from those of their constituent elements.
- Compounds can be separated into simple substances only by chemical methods.

Mixture

- A material which consists of more than one type of substance and which are mixed in any ratio by weight.
- The property of the mixture is the property of its constituent compounds.
- This can be separated into its substances by physical method.

Homogeneous

- A mixture having the same composition throughout (e.g., salt water, alloys).

Heterogeneous

- The mixture in which components are present non-uniformly (e.g., water + sand, blood).

Atom

- Smallest particle of an element which may or may not have independent existence.

Molecule

- Smallest particle of an element or compound which always have independent existence.
- The smallest particle of the substance that has all of the physical and chemical properties of that substance.

Atomicity of Molecules

- The number of atoms present in one molecule is equal to atomicity (e.g., $O_2 = 2$).

Representation of an element

- An element X is represented as ${}_Z^AX$.
- Z = atomic number (No. of Protons), A = mass number (Protons + Neutrons).
- Number of neutrons = $A - Z$.

Cations and Anions

- Electron loss implies $\rightarrow$ cation (+ve charged).
- Electron gain implies $\rightarrow$ anion (-ve charged).

Definition of Isotopes

- Isotopes are the atoms of the same element having the same number of protons but different mass number (and different neutrons).

Definition of Isobars

- Isobars have a different number of protons and neutrons but the same number of nucleons (neutrons + protons).

Definition of Isotones

- Isotones have a different number of protons and mass numbers but the number of neutrons are the same.

Definition of Isodiaphers

- Isodiaphers have the same difference between proton and neutron number ($|n - p| = \text{same}$).

Dalton Atomic Theory

- Matter is made up of very small indivisible particles called atoms.
- All the atoms of a given element are identical in all respect i.e., mass, shape, size, etc.
- Atoms cannot be created or destroyed by any chemical process or physical process.

Limitations of Dalton Atomic Theory

- Indivisibility of atoms (sub-atomic particles p, n, e exist).
- Isotopes and Isobars.
- Gay-Lussac's Law of combining volume.

Laws of Chemical Combination

1. Law of Conservation of mass

- In all physical and chemical changes, the total mass of the reactants is equal to that of products.
- *Limitation:* This Law is not applicable in case of Nuclear Reactions, where mass is converted into energy $E = \Delta mc^2$.

2. Law of constant or definite proportion

- A chemical compound is always made up of the same elements combined together in the same fixed proportion by mass.
- *Limitation:* The law is not applicable if the element exists in different isotopes or for non-stoichiometric compounds.

3. Law of Multiple Proportion

- When two elements combine with each other to form two or more chemical compounds, then different amounts of one element which react with a definite amount of a second element, are in proportion of simple whole numbers.
- *Limitation:* Does not hold well when an element exists in the form of different isotopes or for non-stoichiometric compounds.

4. Law of Reciprocal Proportion

- The ratio of masses of two elements A and B which combine separately with the fixed mass of the third element C is either the same or some simple multiple of the ratio of the masses in which A and B combine directly with each other.

5. Gay Lussac's Law of combining volume

- When gases combine or are produced in a chemical reaction, they do so in a simple ratio by volume, provided that all the gases are at the same temperature and pressure.

Relative Atomic Mass

- $RAM = \frac{\text{mass of one atom of the substance}}{\frac{1}{12} \times \text{mass of one atom of } C-12 \text{ atom}}$
- $RAM = \text{Total number of nucleons (Protons + Neutrons)}$

Atomic Mass Unit

- $1 \text{ u or } 1 \text{ amu} = 1.66 \times 10^{-24} \text{ g or } 1.66 \times 10^{-27} \text{ kg}$

Atomic Mass or Molecular Mass

- $\text{Atomic Mass} = \text{total nucleons in that atom} \times \text{mass of 1 nucleon}$

Mole

- A mole is the amount of the substance that contains as many entities (atom, molecule, ion or other particle) as there are atoms present in exactly 12 g of $C-12$ sample.
- $\text{No. of entities} / N_A = \text{mole of entity}$

Gram Atomic or Gram Molecular Weight

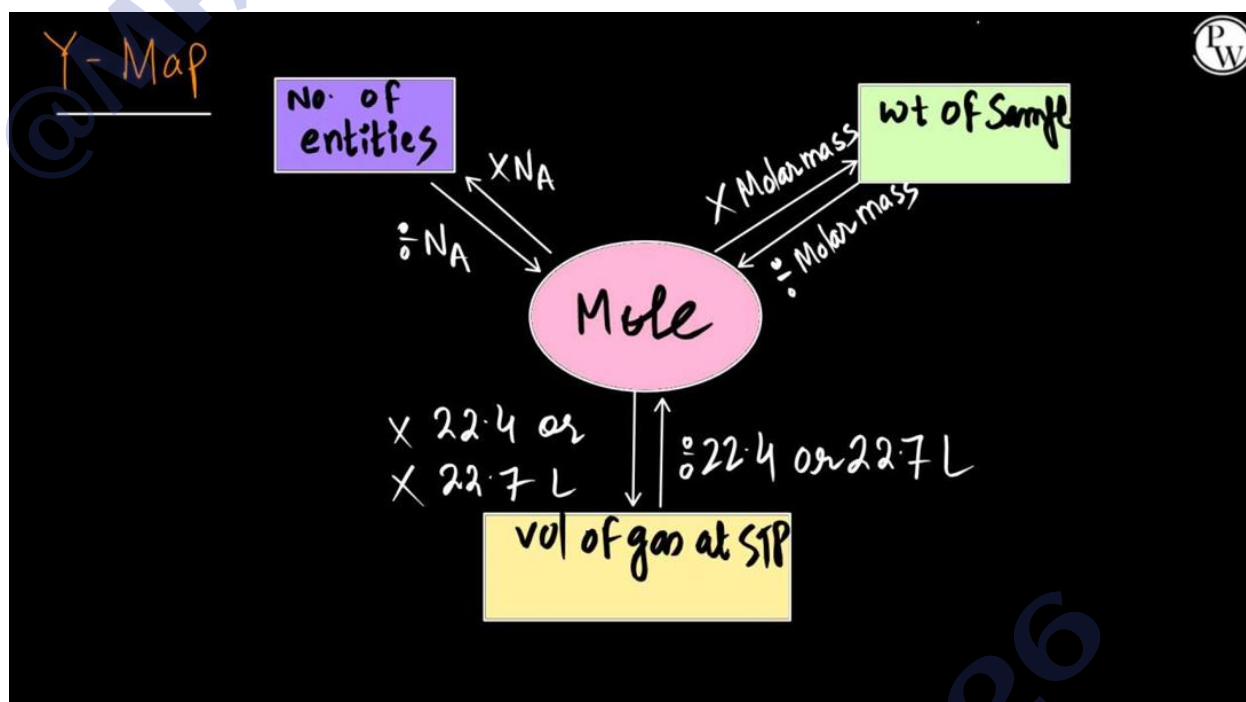
- Weight of one mole of atom = gram atomic weight (weight represented in g not in u).
- Weight of one mole of molecule = gram molecular weight or molar mass (weight represented in g not in u).
- $\text{Moles} = \frac{\text{Total wt of sample in gm}}{\text{Molar mass}}$

Avagadro's Hypothesis

- Equal volume of all gases have equal number of molecules (not atoms) at same temperature and pressure conditions.
- Volume of one mole gas at 0°C and 1 atm = 22.4 L.
- Volume of one mole gas at 0°C and 1 bar = 22.7 L (STP-modern).

Y- Map

- Mole acts as the central link connecting the number of entities, the mass of the sample, and the volume of gas at STP.



Percentage Composition

- Mole of atom in a molecule = mole of molecule $\times$ atomicity of atom.

Empirical Formula

- Molecular Formula:** The formula which gives the exact number of atoms of elements present in one molecule of the substance.
- Empirical Formula:** The formula which gives the simplest ratio of atoms of different elements present in one molecule of the substance.
- $M.F = (E.F)_n$, where $n = \frac{M.F \text{ wt}}{E.F \text{ wt}}$.

Average Atomic Mass

- Average Atomic Mass =
$$\frac{(\% \text{ of isotope 1} \times \text{at. wt. of isotope 1}) + (\% \text{ of isotope 2} \times \text{at. wt. of isotope 2})}{100}$$

Average Molecular Weight

- The average molar mass of the different substances present in the container.
- Average Molecular Weight =
$$\frac{(n_1 M_1 + n_2 M_2 + n_3 M_3 + \dots)}{(n_1 + n_2 + n_3 + \dots)} = \frac{\sum n_i M_i}{\sum n_i}$$

Density

- **Absolute density:** $\text{Absolute density} = \frac{\text{Mass}}{\text{Volume}}$
 - For a gas: $\text{Absolute density} = \frac{PM}{RT}$
- **Specific Gravity:** $\text{Specific gravity} = \frac{\text{Density of substance}}{\text{Density of H}_2\text{O at 4}^\circ\text{C}}$. Numerically, density of a substance in g/cc is equal to specific gravity.
- **Vapour Density (V.D.):** $\text{V.D.} = \frac{\text{Molar Mass}}{2}$.

Stoichiometry

- The essential steps involve writing and balancing the chemical reaction, reading the molar ratios from the balanced equation, and applying the data given in the question.

Limiting Reagent

- The limiting reagent is the reactant that is completely used up in a chemical reaction.
- It determines the maximum amount of product that can be formed.

Percentage Yield

- Percentage Yield is the ratio of the actual amount of product obtained to the theoretical (expected) amount, multiplied by 100.
- $\% \text{ Yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100$